Settling figure methods

theoretical_settling_v2 · settling_velocity_vs_chi
Original methods revision: 20 October 2026

These are analytical calculations in air, separate from AERMOD outputs. No source location, meteorological date range, or field observations are used in the calculation. The associated AERMOD project is near Clemson, South Carolina, and uses 2023 weather; its location and dates do not make these constant-air-property calculations observed settling measurements.

DEFINITION AND EQUATION

The dynamic shape factor χ is the drag of the nonspherical particle divided by the drag of a sphere of equal volume moving at the same speed, with consistent flow-regime and orientation conventions. Chi is a SHAPE factor; diameter separately represents size. Ideal spheres have $\chi = 1$. Larger χ means greater drag and slower settling at fixed volume/density. It is not the Corey shape factor, sphericity, or a generic drag coefficient; numerical values of those quantities cannot be substituted for χ.

The implemented approximate low-Reynolds-number settling expression is:

$$v = (\rho_{particle} - \rho_{air})\, g\, d_{ve}^{\,2}\, C_c(d_{ve}) / (18\, \mu\, \chi)$$

$$C_c = 1 + (2\, \lambda / d_{ve})\, [1.257 + 0.4\, \exp(-0.55\, d_{ve} / \lambda)]$$

where d_{ve} is volume-equivalent sphere diameter in meters. The buoyancy correction is included. Velocity is exported in m/s and plotted in mm/s. Reynolds number is $\rho_{air} \times v \times d_{ve}/\mu$.

Constants, held fixed rather than taken from hourly meteorology:

Fixed property	Value
Gravitational acceleration, g	9.80616 m/s²
Air dynamic viscosity, μ	1.81×10^{-5} Pa s
Air density, ρ_{air}	1.2 kg/m³
Air mean free path, λ	65×10^{-9} m

These approximate near-ambient air properties do not describe every hour of the 2023 meteorology. The constants and spherical slip expression match the verified AERMOD26135 particle-settling implementation used as reference.

The correction matters for the smallest size: C_c is approximately 1.328 at 0.5 micrometers. Applying a sphere's slip correction at d_{ve} and then dividing by a constant continuum χ is a schematic approximation for nonspherical particles. Real effective χ can depend on size/Knudsen number and orientation. These figures do not establish measured fiber or film settling speeds, and they do not alter shape in the AERMOD runs.

SINGLE-PANEL theoretical_settling_v2

The first type of panel from the original theoretical_settling figure is retained: settling velocity versus diameter. The Reynolds-number panel is removed. Diameter range is 0.5-10 micrometers; the sphere curves mark 0.5, 1, 5, and 10 micrometers. Density colors represent 900, 1050, and 1400 kg/m³. Line styles represent χ = 1, 2, and 3. Values 2 and 3 are explicit illustrative sensitivities, not assigned typical values for fibers, fragments, films, or other microplastic morphotypes. The largest Reynolds number in these plotted curves is approximately 0.00284; the low-Reynolds-number Stokes approximation is appropriate for the spherical reference within this restricted size range.

SIMPLE settling_velocity_vs_chi

Volume-equivalent diameter is fixed at 10 micrometers and material density at 1050 kg/m³. The curve varies χ only. Thus v(χ)/v(1) = 1/χ. For χ = 1, 2, and 3, velocities are approximately 3.208, 1.604, and 1.069 mm/s. These are predictions for the stated equivalent-volume surrogate, not observed velocities of particular morphotypes. Literature annotations identify their reference geometry and source. They must not be interpreted as universal microplastic morphotype means.

MORPHOTYPE EVIDENCE AND CAVEATS

The numerical source ledger is morphotype_shape_factors.json; the detailed research record is literature_notes.md. These accompany the figure. Published aerosol reference-shape values can support idealized proxies, while published microplastic studies emphasize aspect ratio, orientation, and flow-regime dependence. An ideal bead may be represented by a sphere; a compact fragment by a cube proxy; a fiber by a specified cylinder proxy. Irregular real fragments and fibers can differ greatly from those proxies. No unsupported χ constant is imputed for film or foam. Hollow/porous particles also require a clearly defined material/effective density and equivalent-volume convention, beyond naming a morphotype.

Reference geometry	χ	Interpretation
Sphere	1.00	Ideal bead/pellet reference
Cube	1.08	Compact-fragment proxy
Average-orientation cylinder, L/D=2	1.10	Specified fiber proxy
Average-orientation cylinder, L/D=5	1.35	Specified fiber proxy
Average-orientation cylinder, L/D=10	1.68	Specified fiber proxy
Film	unassigned	
Foam	unassigned	

MORPHOTYPE EVIDENCE AND CAVEATS — CONTINUED

The table is verified in Particula's Hinds-cited implementation, pinned at commit acf38253e09e8f629bd5e2e93f3aa3a875e5cbd6. Hinds' original textbook table was not independently accessed. It supplies secondary support for reference geometries, not empirically typical environmental microplastic values. The targeted studies inspected did not establish a defensible typical χ per morphotype; this is not proof that no further data exist. Exact source URL and retrieval hashes are supplied in the source ledger.

In particular, measured fiber-to-sphere velocity ratios from large, finite-Reynolds-number microplastic fibers cannot simply be inverted and used as micron-scale dynamic χ values. Values measured in water or reported as Corey/Aschenbrenner shape factors are not substituted here. The research ledger gives actual microplastic study dimensions and conditions wherever relevant, alongside the chosen reference proxies.

Relevant atmospheric microplastic studies include Tatsii et al. (2024), Shape Matters, doi:10.1021/acs.est.3c08209 (full text inspected), and Preston et al. (2023), Effects of Shape and Size on Microplastic Atmospheric Settling Velocity, doi:10.1021/acs.est.3c03671 (abstract inspected; full text unavailable through the accessed sources). Literature_notes.md records the evidence and access limits for these and the other studies checked.

REPRODUCIBILITY

Generator: clemson_amp/scripts/revision_figures.py. CSV files beside the figures contain all sampled curve coordinates, diameter, density, χ, Cunningham correction, velocity, and Reynolds number. The complete figure provenance records exact equation/constants, source ledger hash, script/dependency hashes, rendered file hashes, and whether the figure is analytical or derived from native model output. Original figures remain available unchanged.